

POLICY BRIEFING • FOSSIL FUELS & INDUSTRY • 1750–2024

Global CO₂ Emissions: Who Emits, How Much, and What's Changing

A data-driven overview of global CO₂ emission trends by country, region, and per capita — from the Industrial Revolution to 2024.

Audience: policy analysts, climate negotiators, sustainability researchers

Data: Global Carbon Budget (2025) via Our World in Data

SCOPE

CO₂ only

Fossil fuels & industry (not land-use change)

WINDOW

1750 → 2024

Long-run trends + current snapshot

QUESTION

Shared, unequal

Totals, shares, and per capita responsibilities



Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Global CO₂ emissions over time



SHIFTING SHARES

Europe and the US Fell from 90%+ of Emissions in 1900 to < 1/3 Today

CHART PLACEHOLDER

STACKED AREA (concept)

Regional shares of global CO₂ over time (1900 → 2024): Asia rising (led by China), while Europe + United States decline as a share.

Key points from brief

- ▶ 1900: Europe + US > 90% of global emissions
- ▶ 1950: still > 85%
- ▶ Today: < one-third ; Asia roughly half

Share of global CO₂ emissions by country

Figure: Share of global CO₂ emissions by country

Source: Our World in Data / Global Carbon Budget 2025

Suggested visual: stacked area

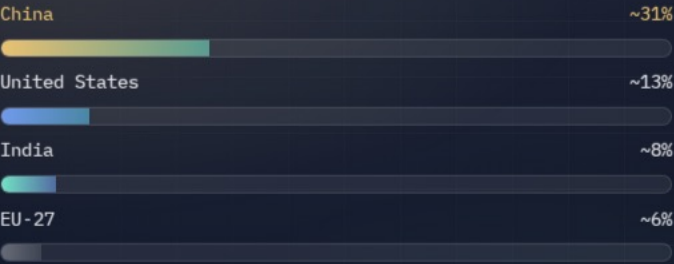
Source: Our World in Data / Global

COUNTRY-LEVEL SHARES (2024)

China Now Emits More Than One-Quarter of Global CO₂

TREEMAP / BAR (PLACEHOLDER)

2024 shares (from brief)



Context: Europe + US historically dominated (>90% in 1900), but Asia now produces roughly half of global emissions.

Suggested visual: treemap or ranked bar

Values must match brief

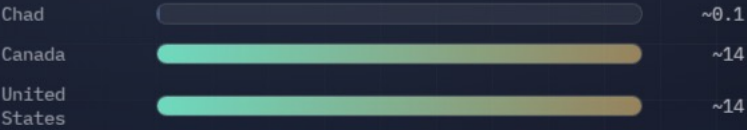
Share of global CO₂ emissions by country

Reference: share chart (OWID) Source: Our World in Data / Global Carbon Budget 2025

PER-CAPITA INEQUALITY (2024)

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

HORIZONTAL BAR CHART (RENDERED)



HEADLINE INEQUALITY

~150x gap

A ~0.1 t/person vs ~14 t/person spread between the lowest and highest large-country emitters.

Values: Chad ~0.1; US ~14; Canada ~14 (t/person) Source: Our World in Data / Global Carbon Budget 2025

INTERPRETATION

- ▶ Per-capita emissions reframe “who emits most” by population.
- ▶ Among wealthy countries, per-capita differences reflect energy systems and policy choices.
- ▶ European countries like France and the UK are much lower than the US despite similar living standards.

CHART OPTION

A world map works well for global context; a ranked bar chart makes the 150x gap explicit.

Suggested alternative: world map choropleth

Note: do not add new values

PER-CAPITA SNAPSHOT (APPROX. 2024)

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India



2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual percentage change in CO₂ emissions, 2024

Figure: Annual % change in CO₂ emissions (2024)

Source: Our World in Data / Global Carbon Budget 2025

READING THE MAP

- ▶ Declines across the US, much of the EU, and the UK (blue shading).
- ▶ Increases in Russia, parts of Southeast Asia, and several African nations (orange/red).
- ▶ Some Central Asian countries (Turkmenistan/Uzbekistan region) show increases exceeding 10% .

OPTIONAL ALTERNATIVE

A diverging bar chart (top movers up/down) can complement the global view — but only with values provided in the brief.

Suggested: map or diverging bar Do not invent country % values

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

COMPARATIVE BAR (RENDERED)

US		~14
EU-27		~6
UK		~4

QUALITATIVE DRIVER (FROM BRIEF)

Among wealthy nations with similar prosperity, per-capita emissions diverge because electricity and energy systems differ: higher shares of **nuclear** and **renewables** can lower per-capita CO₂ relative to fossil-heavy mixes.

Values shown: US ~14, EU-27 ~6, UK ~4 (t/person)

Source: Our World in Data / Global Carbon Budget 2025

WHAT THIS IMPLIES

- ▶ Emissions are not a simple proxy for living standards.
- ▶ Policy + technology choices can decouple prosperity from CO₂.
- ▶ Comparisons should pair totals (shares) with per-capita context.

CHART VARIANT (PLACEHOLDER)

COMPARATIVE BAR (concept)

Show a small set of wealthy countries grouped by “lower” vs “higher” per-capita outcomes, annotated with “energy mix” note. Do not add additional numeric values.

Text and chart slide (qualitative)

No detailed breakdown

CLOSE

Key Takeaways: Responsibility Is Shared but Unequal

- ▶ Global CO₂ emissions from fossil fuels exceed 35 Bt/yr and have not yet peaked.
- ▶ China is the largest annual emitter (~31% of global CO₂), but per-capita emissions (~9 t) remain below the US (~14 t).
- ▶ Historical responsibility has shifted: Europe + the US were >90% of emissions in 1900, but today they are <one-third.
- ▶ Per-capita gaps of roughly 150× persist between the lowest emitters (~0.1 t) and high-emitting wealthy nations (~14–15 t).
- ▶ Energy policy and technology choices can decouple prosperity from emissions (e.g., UK ~4 t vs US ~14 t).

Data: Global Carbon Budget 2025

Via: Our World in Data (Ritchie & Roser)

Coverage: fossil fuels & industry, 1750–2024

